

Financial Time Series Analysis

Lecture 9

- ▶ TA Session Friday April 23, 2021, 11:00am - 12:00pm (EDT)
- ▶ TA Session Monday April 26, 2021, 7:30pm - 8:30pm (EDT)

Outline

- ▶ ARCH Review
- ▶ Properties of ARCH(1)
- ▶ Connection with Stylized Facts
- ▶ Generalization to GARCH
- ▶ Data Example

ARCH Review: 1

- ▶ Recall that ARCH/GARCH models are for the volatility component of a time series model, i.e. they are MDS, hence white noise, e.g. AR(1) + ARCH(1).
- ▶ ARCH is an abbreviation for AutoRegressive Conditional Heteroschedasticity. The conditional and unconditional mean of an ARCH model is 0, the unconditional variance is constant, but the conditional variance depends upon the past variability.
- ▶ $\{a_t, -\infty < t < \infty\}$ is ARCH(1) if

$$a_t = \sigma_t \epsilon_t, \text{ and } \sigma_t = \sqrt{\omega + \alpha a_{t-1}^2},$$

where

$\{\epsilon_t, -\infty < t < \infty\}$ is iid with $E(\epsilon_t) = 0$ and $E(\epsilon_t^2) = 1$.

ARCH Review: 2

- ▶ σ_t^2 is the conditional variance at time t , and letting $\mathcal{F}_t = \sigma\{a_s, s \leq t\}$, σ_t is $\mathcal{F}_{t-1}$ measurable, so $E(\sigma_t|\mathcal{F}_{t-1}) = \omega + \alpha a_{t-1}^2$, a deterministic function given $\mathcal{F}_{t-1}$.
- ▶ $E(a_t) = E(\epsilon_t \sigma_t) = E(E(\epsilon_t \sigma_t | \mathcal{F}_{t-1})) = E(\sigma_t E(\epsilon_t | \mathcal{F}_{t-1})) = E(\epsilon_t \cdot 0) = 0$.
- ▶ $Var(a_t) = E(a_t^2) = E(\epsilon_t^2 \sigma_t^2) = E(\epsilon_t^2 (\omega + \alpha a_{t-1}^2)) = 1(\omega + \alpha E(a_{t-1}^2))$.
- ▶ Assuming weak stationarity (therefore constant variance)
- next slide - we solve for $Var(a_t) = \omega / (1 - \alpha)$.

ARCH Properties: 1

$$\begin{aligned}a_t^2 &= \epsilon_t^2 \sigma_t^2 \\&= \epsilon_t^2 (\omega + \alpha a_{t-1}^2) \\&= \omega \epsilon_t^2 + \alpha a_{t-1}^2 \epsilon_t^2 \\&= \omega \epsilon_t^2 + \alpha \epsilon_t^2 \sigma_{t-1}^2 \epsilon_{t-1}^2 \\&= \omega \epsilon_t^2 + \alpha \epsilon_t^2 \epsilon_{t-1}^2 (\omega + \alpha a_{t-2}^2) \\&= \dots \\&= \omega \sum_{j=0}^n \alpha^j \prod_{i=0}^{t-j} \epsilon_{t-i}^2 + \alpha^{n+1} a_{t-n-1}^2 \prod_{j=0}^n \epsilon_{t-j}^2\end{aligned}$$

If $|\alpha| < 1$, the second term converges to 0, thus

$$a_t^2 = \omega \sum_{j=0}^{\infty} \alpha^j \prod_{i=0}^{t-j} \epsilon_{t-i}^2.$$

Clearly $E(a_t^2) = \omega \sum_{j=0}^{\infty} \alpha^j = \omega / (1 - \alpha)$.

ARCH Properties: 2

- One can express $\{a_t^2\}$ as an AR(1) process because

$$\begin{aligned}a_t^2 &= \sigma_t^2 + a_t^2 - \sigma_t^2 \\&= \omega + \alpha a_{t-1}^2 + \sigma_t^2 \epsilon_t^2 - \sigma_t^2 \\&= \omega + \alpha a_{t-1}^2 + \sigma_t^2 (\epsilon_t^2 - 1) \\&= \omega + \alpha a_{t-1}^2 + \nu_t.\end{aligned}$$

Thus $\{a_t^2\}$ is an AR(1) process with noise process $\{\nu_t\}$, where $\nu_t = \sigma_t^2(\epsilon_t^2 - 1)$.

- We show that $\{\nu_t\}$ is a white noise process assuming $E(\epsilon_t^4) < \infty$.

ARCH Properties: 3

$$a_t^2 = \omega + \alpha a_{t-1}^2 + \nu_t,$$

where $\nu_t = \sigma_t^2(\epsilon_t^2 - 1)$.

- ▶ $E(\nu_t) = E(E(\sigma_t^2(\epsilon_t^2 - 1)|\mathcal{F}_{t-1})) = E(\sigma_t^2 E(\epsilon_t^2 - 1|\mathcal{F}_{t-1})) = E(\sigma_t^2 \cdot 0) = 0$.
- ▶ $\gamma_\nu(h) = E(\nu_{t+h}\nu_t) = E(\sigma_{t+h}^2 \sigma_t^2 (\epsilon_{t+h}^2 - 1)(\epsilon_t^2 - 1)) = E(\epsilon_{t+h}^2 - 1)E((\epsilon_t^2 - 1)\sigma_{t+h}^2 \sigma_t^2) = 0, h \geq 1$. Thus $\{\nu_t\}$ a white noise process.
- ▶ We only need to calculate $Var(\nu_t) = E(\sigma_t^4(\epsilon_t^2 - 1)^2) = E(\sigma_t^4)E(\epsilon_t^4 - 2 \cdot 1 + 1) = 2E(\sigma_t^4)$ assuming ϵ_t has a $N(0,1)$ distribution.

ARCH Properties: 4

Because a_t^2 is an AR(1) process, we know

$$E(a_t^2) = \omega / (1 - \alpha),$$

$$\text{Var}(a_t^2) = \text{Var}(\nu_t) / (1 - \alpha^2) = 2E(\sigma_t^4) / (1 - \alpha^2).$$

To calculate $E(\sigma_t^4)$,

$$\begin{aligned} E(\sigma_t^4) &= E((\omega + \alpha a_{t-1}^2)^2) \\ &= \omega^2 + 2\alpha\omega(\omega / (1 - \alpha)) + \alpha^2(\text{Var}(a_{t-1}^2) + E(a_{t-1}^2)^2) \\ &= \omega^2 + \frac{2\omega^2\alpha}{(1 - \alpha)} + \frac{2E(\sigma_t^4)\alpha^2}{1 - \alpha^2} + \frac{\omega^2\alpha^2}{(1 - \alpha)^2} \end{aligned}$$

Solving for $E(\sigma_t^4)$, we find

ARCH Properties: 5

$$E(\sigma_t^4) \frac{1 - 3\alpha^2}{1 + \alpha} = \frac{\omega^2}{1 - \alpha}.$$

- ▶ This requires $\alpha^2 < 1/3$ or $0 \leq \alpha < .577$, assuming that ϵ_t has a $N(0,1)$ distribution.
- ▶ Solving the above equation for $E(\sigma_t^4)$:

$$E(\sigma_t^4) = \frac{\omega^2(1 + \alpha)}{(1 - 3\alpha^2)(1 - \alpha)}.$$

- ▶ It follows that

$$\begin{aligned} E(a_t^4) &= E(\sigma_t^4 \epsilon_t^4) = 3E(\sigma_t^4) \\ &= 3 \frac{\omega^2(1 + \alpha)}{(1 - \alpha)(1 - 3\alpha^2)} \\ &= 3 \left(\frac{\omega}{(1 - \alpha)} \right)^2 \frac{1 - \alpha^2}{1 - 3\alpha^2} \\ &= 3(E(a_t^2))^2 \frac{1 - \alpha^2}{1 - 3\alpha^2} \end{aligned}$$

ARCH Properties: 6

- ▶ Since the last factor is larger than 1, we conclude that $E(a_t^4)/(E(a_t^2)^2)$ is larger than 3, which is indicative of heavier tails than the $N(0,1)$ distribution.
- ▶ Note also that the AR(1) structure of $\{a_t^2\}$,

$$a_t^2 = \omega + \alpha a_{t-1}^2 + \nu_t,$$

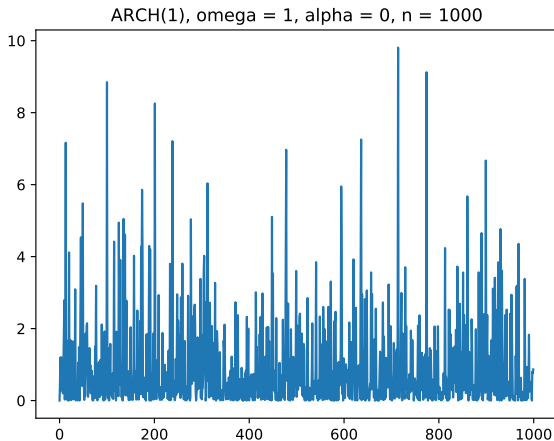
where $\{\nu_t\}$ is white noise allows us to conclude that

$$\rho_{a^2}(h) = \alpha^h \geq 0.$$

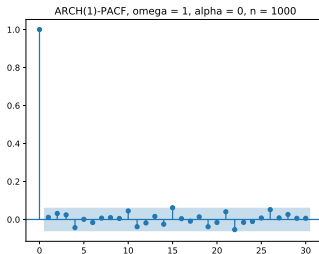
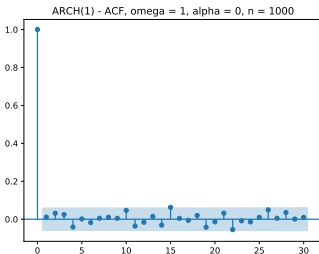
Thus the ACF of $\{a_t^2\}$ is non-negative.

- ▶ That a_t^2 and a_{t+h}^2 are positively correlated (especially for small values of h) is known as the **ARCH Effect**.

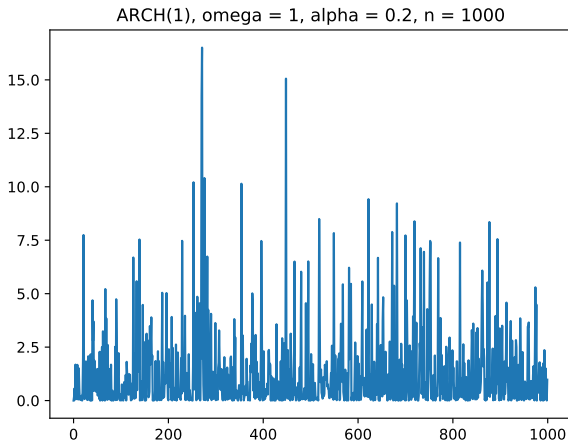
$$a_t^2, \omega = 1, \alpha = 0, n = 1000$$



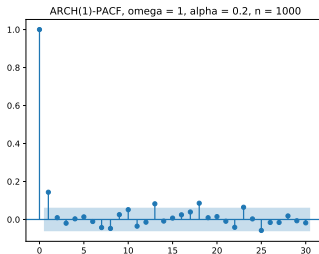
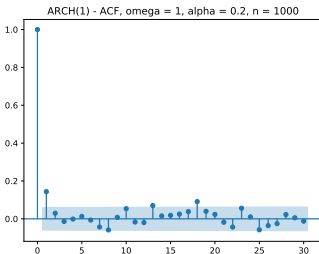
$$a_t^2, \omega = 1, \alpha = 0, n = 1000$$



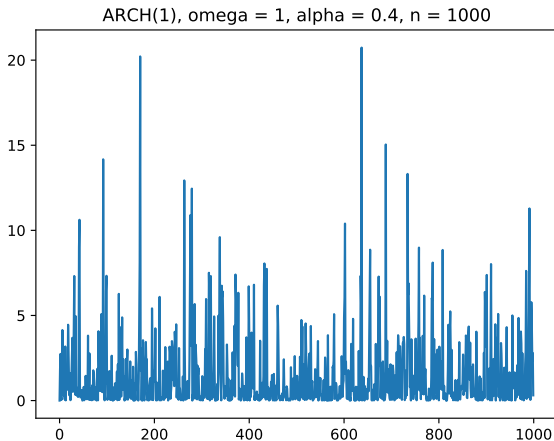
$$a_t^2, \omega = 1, \alpha = .2, n = 1000$$



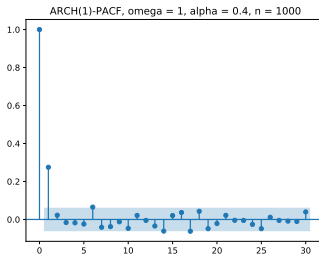
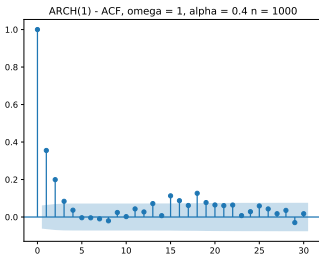
$$a_t^2, \omega = 1, \alpha = .2, n = 1000$$



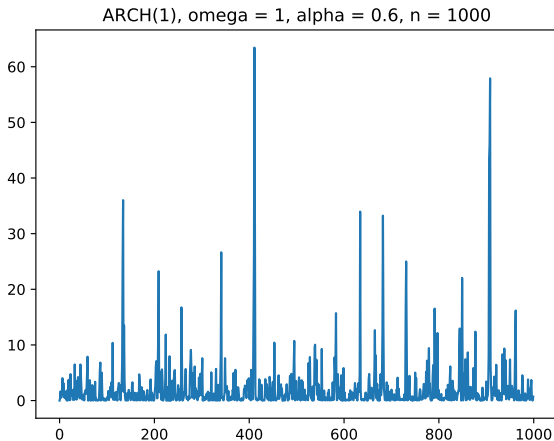
$$a_t^2, \omega = 1, \alpha = .4, n = 1000$$



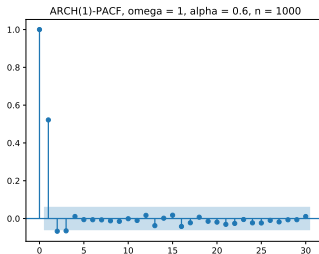
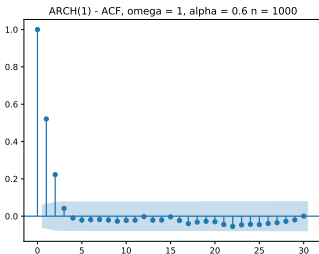
$$a_t^2, \omega = 1, \alpha = .4, n = 1000$$



$$a_t^2, \omega = 1, \alpha = .6, n = 1000$$



$$a_t^2, \omega = 1, \alpha = .6, n = 1000$$



ARCH and Stylized Facts

The ARCH(1) model does support some of the stylized facts:

- ▶ (Absence of Autocorrelation) - ARCH noise models are MDS and are white noise, so $Cov(a_t, a_{t+h}) = 0$, $h \neq 0$; however, $Cov(a_t^2, a_{t+h}^2) > 0$ (ARCH Effect). This implies volatility clustering.
- ▶ Prices are non-stationary, but returns are weakly stationary.
- ▶ Distribution of returns have heavier tails than the normal distribution.
- ▶ ARCH models do not support asymmetry or the leverage effect. $a_t = \epsilon_t \sigma_t = \epsilon_t \sqrt{\omega + \alpha a_{t-1}^2}$. Since the returns depend on the square of a_t , there is no difference between positive and negative values of a_{t-1} . There are variations of the ARCH/GARCH model that overcome this drawback.

Weaknesses of ARCH Model

R. Tsay (2002) notes some weaknesses of the ARCH Model:

- ▶ The model assumes that positive and negative shocks have the same effects on volatility because the model assumes that the next shock depends only on the square of the previous shock(s). The stylized facts indicate that the returns respond differently to positive and negative shocks.
- ▶ The model is restrictive. With $\{\epsilon_t\}$ assumed to be iid $N(0,1)$, $E(a_t^4)$ exists only if $\alpha^2 < 1/3$.
- ▶ The ARCH model does not provide any insight into understanding the source of volatility. It only provides a mechanical way to describe the behavior of the conditional variance. It gives no indication about what causes such behavior to occur. Furthermore, it applies to single stocks and does not account for the effects that are occurring across market segments.

AR(1)/ARCH(1)

We can combine the ARCH noise process with the AR time series model to achieve a process both of whose conditional mean and variance depend on the past.

- ▶ Suppose $\{a_t, t \in \mathcal{Z}\}$ is an ARCH(1) noise process where

$$a_t = \sqrt{\omega + \alpha a_t^2} \epsilon_t,$$

where $\{\epsilon_t, t \in \mathcal{Z}\}$ consists of i.i.d. $N(0,1)$ random variables, $\omega > 0$, $0 \leq \alpha < 1$, $|\beta| < 1$, and define

$$X_t = \mu + \beta(X_{t-1} - \mu) + a_t.$$

- ▶ $\rho_X(h) = \beta^{|h|}$ and $\rho_{a^2} = \alpha^{|h|}$.
- ▶ This model has non-constant conditional mean and variance.

ARCH(p)

- ▶ The order of the ARCH process can easily be increased from 1 to $p > 1$. Let

$$a_t = \sigma_t \epsilon_t,$$

where

$$\sigma_t = \sqrt{\omega + \sum_{i=1}^p \alpha_i a_{t-i}^2}.$$

- ▶ The ARCH(p) process is uncorrelated and has constant conditional and unconditional mean. Its unconditional variance is also constant, but the conditional variance is nonconstant.
- ▶ The ACF of a_t^2 is the same as the ACF of an AR(p) process.

From ARCH to GARCH

- ▶ The original ARCH model was created by R. Engle in 1982. It was generalized to GARCH (generalized autoregressive conditional heteroskedasticity) by Tim Bollerslev in 1986.
- ▶ The GARCH model allows the values of the X process to impact the evolution of the conditional variance, σ_t .
- ▶ The GARCH(p,q) model is given by $\omega \geq 0, \alpha_i \geq 0, \beta_j > 0$:

$$\begin{aligned}a_t &= \sigma_t \epsilon_t, \{\epsilon_t\} \text{ is } N(0, 1) \\ \sigma_t^2 &= \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2\end{aligned}$$

- ▶ There are many generalizations of GARCH(p,q) including EGARCH, GARCH with leverage, IGARCH, Multivariate GARCH, etc.

GARCH(1,1):1

Consider a GARCH(1,1) model with $a_t = \sigma_t \epsilon_t$, and $\{\epsilon_t, t \geq 0\}$ (i.i.d. white noise), a

$$\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2,$$

an ARMA(1,1) process but with a weak white noise error process.

$$\begin{aligned} E(a_t^2 | \mathcal{F}_{t-1}) &= E(\sigma_t^2 \epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= E(\sigma_t^2 | \mathcal{F}_{t-1}) \cdot 1 \\ &= E(\omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2 | \mathcal{F}_{t-1}) \\ &= \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2 \end{aligned}$$

GARCH(1,1):2

- ▶ Define $\eta_t = a_t^2 - \sigma_t^2$
- ▶ $E(\eta_t | \mathcal{F}_{t-1}) = E(a_t^2 - \sigma_t^2 | \mathcal{F}_{t-1}) = 0$.
- ▶ For $s < t$, $E(\eta_s \eta_t) = E(\eta_s E(\eta_t | \mathcal{F}_s)) = 0$, by conditioning on $\mathcal{F}_{t-1}$.
- ▶ Consequently the $\{\eta_t, t \geq 0\}$ process is weak white noise.
- ▶ Some algebra gives

$$\sigma_t^2 = \omega + (\alpha + \beta)a_{t-1}^2 - \beta\eta_{t-1}.$$

GARCH(1,1):3

Let $c = \omega / (1 - (\alpha + \beta))$

$$\begin{aligned}a_t^2 - c &= -c + \sigma_t^2 + \eta_t \\&= -c + \omega + (\alpha + \beta)a_{t-1}^2 - \beta\eta_{t-1} + \eta_t \\&= (\alpha + \beta)(a_{t-1}^2 - c) - \beta\eta_{t-1} + \eta_t\end{aligned}$$

- ▶ It follows that $\{a_t^2, t \geq 0\}$ is an ARMA(1,1) process with mean c and AR(1) coefficient $(\alpha + \beta)$.



$$\rho_{a^2}(h) = (\alpha + \beta)^h \frac{\alpha(1 - \alpha\beta - \beta^2)}{1 - 2\alpha\beta - \beta^2}$$

GARCH(1,1):4

- ▶ It follows that $\{a_t^2, t \geq 0\}$ is an ARMA(1,1) process with mean ν and AR(1) coefficient $\phi_1 = \alpha_1 + \beta_1$, and MA(1) coefficient $\theta_1 = -\beta_1$.



$$\rho_{a^2}(h) = (\alpha_1 + \beta_1)^h \frac{\alpha_1(1 - \alpha_1\beta_1 - \beta_1^2)}{1 - 2\alpha_1\beta_1 - \beta_1^2}$$

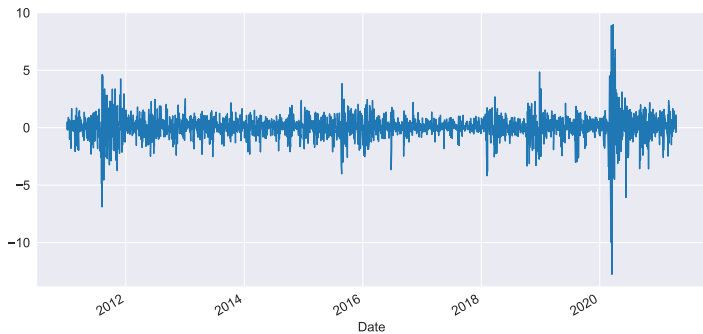
Fitting ARCH to S&P 500 Data: 1

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import datetime as dt
import pandas_datareader as pdr
start = dt.datetime(2011, 1, 1)
end = dt.datetime(2021, 4, 16)
sandp = pdr.get_data_yahoo('GSPC', start, end)
sandp_ac = sandp['Adj Close']
logret = 100*(np.log(sandp_ac/sandp_ac.shift(1)).dropna())
```

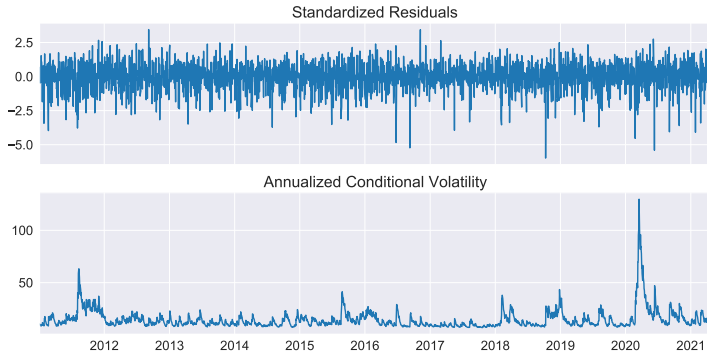
Fitting ARCH to S&P 500 Data: 2

```
figure = logret.plot()  
plt.show()  
from arch import arch_model  
am = arch_model(logret)  
res = am.fit(update_freq=5)  
print(res.summary())  
fig = res.plot(annualize='D')  
plt.show()
```

S&P 500



S&P 500



GARCH(p,q):1

- ▶ We have written the GARCH(1,1) as an ARMA(1,1) model with an MDS noise process. This can be generalized to the GARCH(p,q)

$$a_t = \sigma_t \epsilon_t,$$
$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

where we assume $\{\epsilon_t\}$ are iid with mean 0 and variance 1.

- ▶ If we define $\alpha_i = 0$ if $i > p$ and $\beta_j = 0$ if $j \geq q$, then letting $\eta_t = a_t^2 - \sigma_t^2$ or $a_t^2 = \sigma_t^2 + \eta_t$, we find:

$$\begin{aligned} a_t^2 &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j (a_t^2 - \sigma_t^2) \\ &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j \eta_t \end{aligned}$$

GARCH(p,q):2

- ▶ η_t is a MDS, because $E(a_t^2|\mathcal{F}_{t-1}) = \sigma_t^2$, so
 $E(\eta_t|\mathcal{F}_{t-1}) = E(a_t^2|\mathcal{F}_{t-1}) - \sigma_{t-1}^2 = \sigma_t^2 - \sigma_{t-1}^2 = 0$.
- ▶ Consequently, $\{a_t^2\}$ is an ARMA(p,q) with AR coefficients $\alpha_j + \beta_j$ and MA coefficients β_j .
- ▶ The variance of η_t is constant assuming $E(\epsilon_t^4) < \infty$
Consequently, η_t is a white noise process.
- ▶ Because the error process is a white noise process, and all of our previous results hold.

GARCH(p,q):3

- ▶ To ensure that the $\{a_t^2\}$ process is weakly stationary, we need the roots of the polynomial

$$1 - \sum_{j=1}^{\max p, q} (\alpha_j + \beta_j) z^j,$$

to lie outside the unit circle.

- ▶ We usually assume that all of the α_i and β_j are positive, the weak stationary condition becomes

$$\sum_{i=1}^p \alpha_i + \sum_{j=1}^q \beta_j < 1.$$

- ▶ If this condition holds, then

$$E(a_t^2) = \omega / (1 - \sum_{i=1}^p \alpha_i - \sum_{j=1}^q \beta_j).$$

GARCH Forecasting: 1

- ▶ Suppose we wish to forecast X_{t+k} for some $k \geq 1$ where the $\{X_t\}$ process is $\text{ARMA}(p_m, q_m) + \text{GARCH}(p_v, q_v)$.
- ▶ The $\text{GARCH}(p_v, q_v)$ noise process is white noise, and we developed ARMA forecasting assuming white noise errors. So the point forecasts of X_{t+k} will be the same as if we assumed the errors were iid Gaussian.
- ▶ What is different is that the GARCH noise process will impact the variability of the forecast, specifically the forecast intervals.

GARCH Forecasting: 2

- ▶ We showed that a $\text{GARCH}(p_v, q_v)$ can be represented as an ARMA process with the MDS.
- ▶ Consequently, we can use the ARMA forecasting techniques on the GARCH model.
- ▶ We will first review ARMA forecasting and then apply it to GARCH forecasting.

GARCH Forecasting: 3

- ▶ We showed that the GARCH(1,1) can be written as an ARMA process with MDS noise, hence $WN(0, \sigma_a^2)$. Let us review how to forecast ARMA(p,q) processes:

$$X_t = \phi_0 + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t.$$

- ▶ Recall that for forecasting a future value, say $\hat{X}_{t+k}$, we use the defining model above and forecast future values of the process, $(\hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+k})$ sequentially.
- ▶ At each stage, we plug in the current data $(X_t, X_{t-1}, \dots, X_{t-k})$ as called for by the model, and we plug in the estimates of the unknown parameters.

GARCH Forecasting: 4

- ▶ This leaves the ϵ terms in the equation. For current and previous values of ϵ , we plug in $(\hat{\epsilon}_t, \hat{\epsilon}_{t-1}, \dots, \hat{\epsilon}_{t-q})$ going backward in time, and we plug in 0 going forward, i.e. $\hat{\epsilon}_{t+i} = 0$ for $i \geq 1$.
- ▶ For a GARCH(1,1) volatility process with parameters $\alpha_1 > 0$ and $\beta_1 > 0$ with $\alpha_1 + \beta_1 < 1$,

$$a_t^2 = \epsilon_t^2(\omega + (\alpha_1 + \beta_1)a_{t-1}^2 - \beta_1\eta_t).$$

- ▶ Let the constant unconditional variance be σ^2 . Taking expectations across the equation, we find

$$\sigma^2 = \omega / (1 - \alpha_1 - \beta_1).$$

GARCH Forecasting: 5

- ▶ Suppose that the current time is t and we want to forecast $\hat{\sigma}_{t+k}^2$, the k -step ahead forecast of the volatility given $\mathcal{F}_t$
- ▶ Consider the one-step ahead forecast, $\hat{\sigma}_{t+1}^2$. We have

$$\sigma_{t+1}^2 = \omega + \alpha_1 a_t^2 + \beta_1 \sigma_t^2.$$

The right-side is $\mathcal{F}_t$ measurable, so the above equation provides the one-step-ahead forecast, although estimates must be substituted for the unknown parameters, $\omega, \alpha_1, \beta_1$.

- ▶ For multi-step forecasts, $k \geq 1$, recall that $a_t^2 = \epsilon_t \sigma_t^2$. So the volatility equation can be rewritten as

$$\sigma_{t+1}^2 = \omega + \alpha_1 \epsilon_t \sigma_t^2 + \beta_1 \sigma_t^2$$

GARCH Forecasting: 6

- ▶ or rearranging

$$\sigma_{t+1}^2 = \omega + (\alpha_1 + \beta_1)\sigma_t^2 + \alpha_1\sigma_t^2(\epsilon_t^2 - 1).$$

- ▶ The forecast at time $t + 2$ is $E(\hat{\sigma}_{t+2}^2 | \mathcal{F}_t)$.
- ▶ Changing $t + 1$ to $t + 2$ in the above equation and taking $E(\cdot | \mathcal{F}_t)$ and noting $E(\epsilon_{t+1} - 1 | \mathcal{F}_t) = 0$, we find

$$\hat{\sigma}_{t+2}^2 = \omega + (\alpha_1 + \beta_1)\hat{\sigma}_{t+1}^2.$$

This generalizes to

GARCH Forecasting: 7

$$\hat{\sigma}_{t+k}^2 = \omega + (\alpha_1 + \beta_1)\hat{\sigma}_{t+k-1}.$$

We let $\lambda = \alpha_1 + \beta_1$ and iterate backward:

$$\begin{aligned}\hat{\sigma}_{t+k}^2 &= \omega + \lambda\hat{\sigma}_{t+k-1}^2 \\ &= \omega(1 + \lambda) + \lambda^2\hat{\sigma}_{t+k-2}^2 \\ &= \dots \\ &= \omega(1 + \lambda + \dots + \lambda^{k-2}) + \lambda^{k-1}\hat{\sigma}_{t+1}^2\end{aligned}$$

As $k \rightarrow \infty$, assuming $\lambda < 1$

$$\hat{\sigma}_{t+k}^2 \rightarrow \sigma^2 = \omega/(1 - \lambda).$$

The **half-life** of the volatility difference is approximately

$$\lambda^T = \frac{1}{2}, \text{ so } T \approx -\frac{\log 2}{\log \lambda} = -\frac{.693}{\log \lambda}$$